

(Please write your Enrollment Number)

Enrollment No. _____

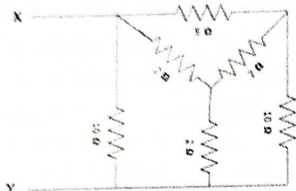
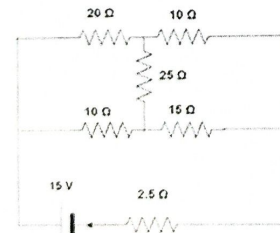
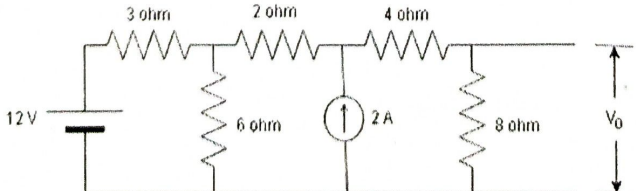
RE-MID TERM EXAMINATION

(Course Name: B. Tech. (ECE/CSE/MAE/DMAM) (Semester: 1st)

(NOVEMBER 2024) OFFLINE mode

Subject Code: BEC-101	Subject: Basic Electrical and Electronics Engineering
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1	(2.5*4=10)	CO Mapping
a) Find the equivalent resistance between X and Y. 		CO1
b) Three resistances R, 2R and 3R are connected in delta. Determine the resistances for an equivalent star connection?		CO2
c) Differentiate between Q-factor and power factor of a series resonant circuit with relevant equations.		CO1
d) An alternating voltage $E=300 \sin 314t$ is applied to a device which offers an ohmic resistance of 30Ω to the flow of current in one direction, while preventing the flow of current in opposite direction. Calculate Form Factor and Peak Factor?		CO2
UNIT I		CO Mapping
Q2 Attempt any two parts	(5*2=10)	
Determine the total current I in the given circuit. 		CO1
State and Prove Maximum Power Transfer Theorem.		CO2
Calculate the voltage V_0 by Thevenin's Theorem. 		CO1
UNIT II		CO Mapping

Q3	Attempt any two parts	(5*2=10)	
a)	A series resonance network consisting of a resistor of 30Ω, a capacitor of $2\mu\text{F}$ and an inductor of 20mH is connected across a sinusoidal supply voltage which has a constant output of 9 volts at all frequencies. Calculate (i) The resonant frequency. (ii) The current at resonance. (iii) The voltage across the inductor and capacitor at resonance. (iv) The quality factor and the bandwidth of the circuit. (v) The upper and lower 3-dB frequencies. (vi) Current Waveform.		CO1
b)	A parallel resonance network consisting of a resistor of 60Ω, a capacitor of $120\mu\text{F}$ and an inductor of 200mH is connected across a sinusoidal supply voltage which has a constant output of 100 volts at all frequencies. Calculate (i) The resonant frequency (ii) The quality factor (iii) The bandwidth of the circuit (iv) The circuit current at resonance and (v) Current magnification.		CO1
c)	Explain R-L-C Series circuit with phasor diagram.		CO2